Mar. 10

- Thm 6.1 (Cayley's Theorem): every group is isomorphic to a group of permutations

i.e every group is isomorphic to a subgroup of a symmetric group

Let G be a finite group. Then $\exists H \leq S_n$ for some n s.t $G \cong H$.

How we usually use this thm.

$\varphi: G \to H$ s.t $\varphi(ab) = \varphi(a)\varphi(b)$ $\forall a,b \in G$ and φ is a bijection

in G — homomorphism — in H — isomorphism

- Example of homomorphism but not isomorphism

$G = \langle r \rangle$ $\quad \varphi \quad$ $H = S_4$

$r \longmapsto (124)$

$r \longmapsto (142)$

$r^3 = e \longmapsto id = (1)(2)(4)$

function that preserves group op.

φ is a homomorphism, but not isomorphism since it is not onto

homomorphism that additionally is bijective, iso is just a stronger condition for a homomorphism

Aside: "representation theory" is the study of representing groups as other groups

any element of S_4 $\qquad \langle r \rangle$

$\alpha \longmapsto e$

homomorphism, but neither one-to-one nor onto, so not isomorphic. This is often called the trivial homomorphism

Proof of Cayley's Thm: Let G be a group. Idea: need permutations to come from somewhere, so we will permute elements of G with elements of G

for $a \in G$, define $T_a: G \to G$ by $T_a(x) = ax$. i.e T_a is group operationing on the left

We claim T_a is a permutation.

Indeed, $T_a: G \to G$ since G is closed. It is a function ✓

To see T_a is bijective, we note $T_{a^{-1}}$ is T_a^{-1}.

That is, $(T_{a^{-1}} \circ T_a)(x) = a^{-1}ax = x$ and $(T_a \circ T_{a^{-1}})(x) = aa^{-1}x = x$. So, T_a is bijective.

Define $LRR(G) = \{T_a : a \in G\}$. This is called the left regular representation of G.

We claim $LRR(G)$ is a group under function composition.

Closure: $(T_a \circ T_b)(x) = abx = T_{ab}(x)$ ✓

Identity: Because $T_e(x) = ex = x$, We check $T_e \circ T_a = T_a = T_a \circ T_e$. ✓

Associativity: $(T_a \circ (T_b \circ T_c))(x) = T_a(T_b(T_c x)) = T_a(bcx) = abcx$

$((T_a \circ T_b) \circ T_c)(x) = (T_a \circ T_b)(cx) = abcx$. ✓

Inverses: See above in proof of bijectivity

note: a porism is a direct consequence of a proof

So, $LRR(G)$ is a group under function composition.

All that remains is to construct an isomorphism between G and $LRR(G)$. Define $\Phi: G \to LRR(G)$ by $\Phi(x) = T_x$. We claim Φ is an isomorphism.

Indeed, $\Phi(ab) = T_{ab} = T_a \circ T_b = \Phi_a \circ \Phi_b$. So Φ is a homomorphism. To see it's a bijection, we note Φ^{-1} is given by $\Phi^{-1}(T_x) = x$. Checking this is a two-sided inverse is left as an exercise to the reader. $\square$

- Note we mapped G into $S_{|G|}$. e.g $LRR(S_5) \leq S_{|S_5|} = S_{120}$. $|S_{120}| = 120!$ is a huge number :o!

Ex. $LRR(\mathbb{Z}/3\mathbb{Z}) < S_3$

$\mathbb{Z}/3\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}\}$, label a, b, c respectively

So $\mathbb{Z}/3\mathbb{Z} \cong \langle abc \rangle$

$T_{\bar{0}} = id$, $T_{\bar{1}}: \begin{array}{l} \bar{0} \mapsto \bar{1} \\ \bar{1} \mapsto \bar{2} \\ \bar{2} \mapsto \bar{0} \end{array} \Rightarrow T_{\bar{1}} = (abc)$, $T_{\bar{2}} = (acb)$

$LRR(\mathbb{Z}/3\mathbb{Z}) = \langle abc \rangle$

Chapter 7: Cosets and Lagrange's Theorem

- **Def.** Let G be a group and H a subgroup of G. for any $a \in G$, let $aH = \{ah : h \in H\}$. We call aH the left coset of H determined by a in G. The set of such sets is called the left cosets of H in G. Likewise, $Ha = \{ha : h \in H\}$ is the right coset of H in G determined by a. $\{Ha : a \in G\}$ is the set of right cosets of H in G.
  - In these cases, we call a a coset representative
  - $|aH|$ is the order of aH, i.e, the number of elements in aH.
  - Note that cosets are either left or right, only both/same if group is Abelian

Ex $\langle 4 \rangle < \mathbb{Z}$. Compute $1 + \langle 4 \rangle$

$\langle 4 \rangle = \{0, \pm 4, \pm 8, \pm 12, \ldots\}$ alt where $a = 1$, $H = \langle 4 \rangle$, and group operation is $+$

$1 + \langle 4 \rangle = \{1 + h : h \in \langle 4 \rangle\}$

$= \{1+4, 1+8, 1+12, 1-4, 1-8 \ldots\}$ notice

$= \{\ldots, -7, -3, 1, 5, 9, 13, \ldots\} = \{n \in \mathbb{Z} : n \equiv 1 \bmod 4\} = \bar{1}_4$

Ex. Let $H < G$ be $H = \langle (123) \rangle$, $G = S_3$

- $eH = e\langle (123) \rangle = \{(123), (132), e\}$ left coset of $\langle (123) \rangle$ in S_3 determined by e
- $(12)H = (12)\langle (123) \rangle = \{(12)(123), (12)(132), (12)e\} = \{(23), (13), (12)\}$
- $(123)H = (123)\langle (123) \rangle = \{(123)(123), (123)(132), (123)e\} = \{(132), e, (123)\}$

notice these are the same elements as $\langle (123) \rangle$, just permuted, instance of Cayley's Thm

same coset, different representatives, like how $\bar{1} = \bar{5}$ are the same mod 4